IIT JAM Mathematical Statistics: Previous-Year Paper Analysis

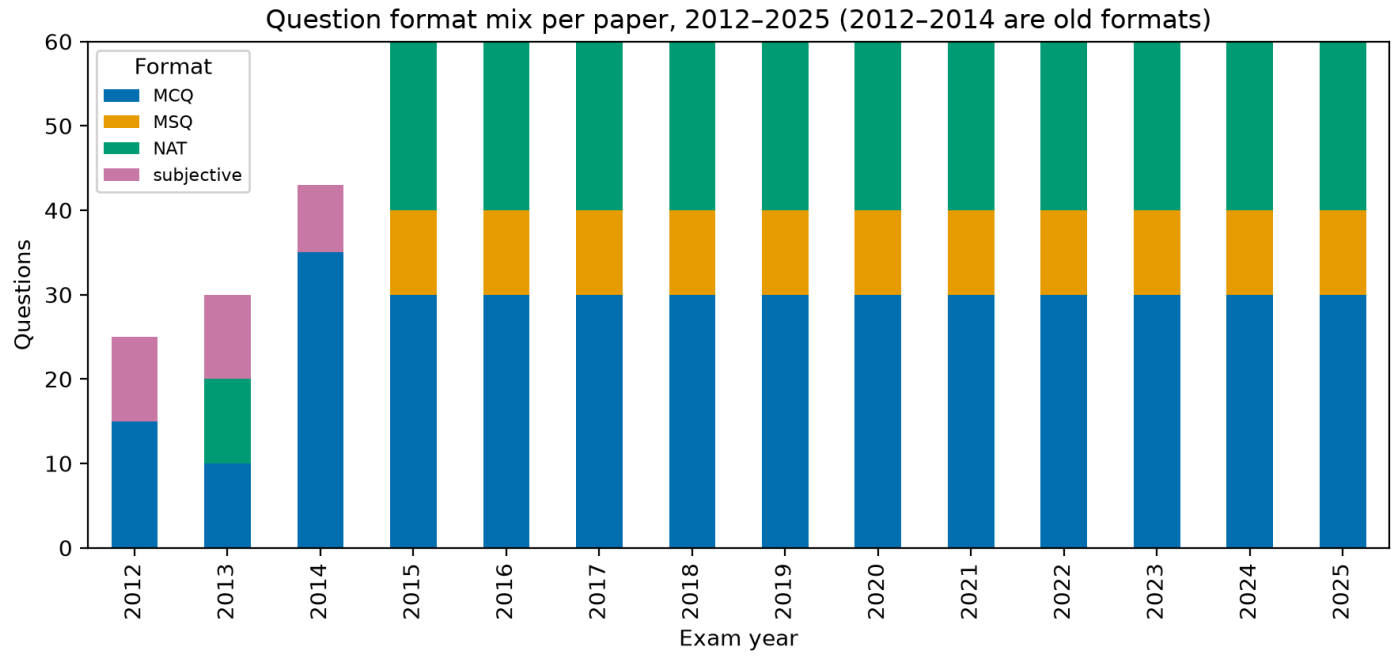
Scope: 11 current-format papers (2015–2025, 60 questions / 100 marks each), 660 questions. Old-format papers 2012–2014 are reported separately. Taxonomy: official JAM 2026 MS syllabus (plus a pre-2026 Differential Equations topic). Evidence tags: strong = seen in nearly every paper; moderate = consistent but small or recent; weak = few observations.

1. Summary

- **Statistics carries 63.9% of marks, Mathematics 36.1%.** strong
- **Univariate Distributions, Estimation and Multivariate Distributions** are the heaviest topics (14.9%, 13.9%, 11.2% of a paper on average); together **40.0%**. strong
- Every main syllabus topic (except the dropped Differential Equations) appears in at least 10 of 11 papers: **no topic is skippable**. strong
- **The paper is shifting** from calculus towards distribution theory: comparing 2015–2018 with 2023–2025, Integral Calculus fell 10.2% → 5.3% while Multivariate Distributions rose 9.2% → 14.3%. moderate
- **43.6%** of questions (288/660) reuse the problem structure of an earlier question with new numbers or set-up; 0 are word-for-word repeats. Drilling past papers by structure pays off. strong
- Differential Equations appear only in 2012–2016 and not since; they are not in the 2026 syllabus. strong
- 11 subtopics of the 2026 syllabus were **never tested** in 2015–2025 (Descriptive Statistics, Nonparametric Methods, Stochastic Processes, among others). They are likely new; treat them as unknown, not low priority.

2. Examination overview

Format since 2015: 30 MCQ (Section A), 10 MSQ (B), 20 NAT (C); 10+20 one/two-mark MCQs, two-mark MSQs, 10+10 one/two-mark NATs. Question styles: numerical 47.1%, application 23.9%, conceptual 22.4%, everything else 6.5% (subjective labels).

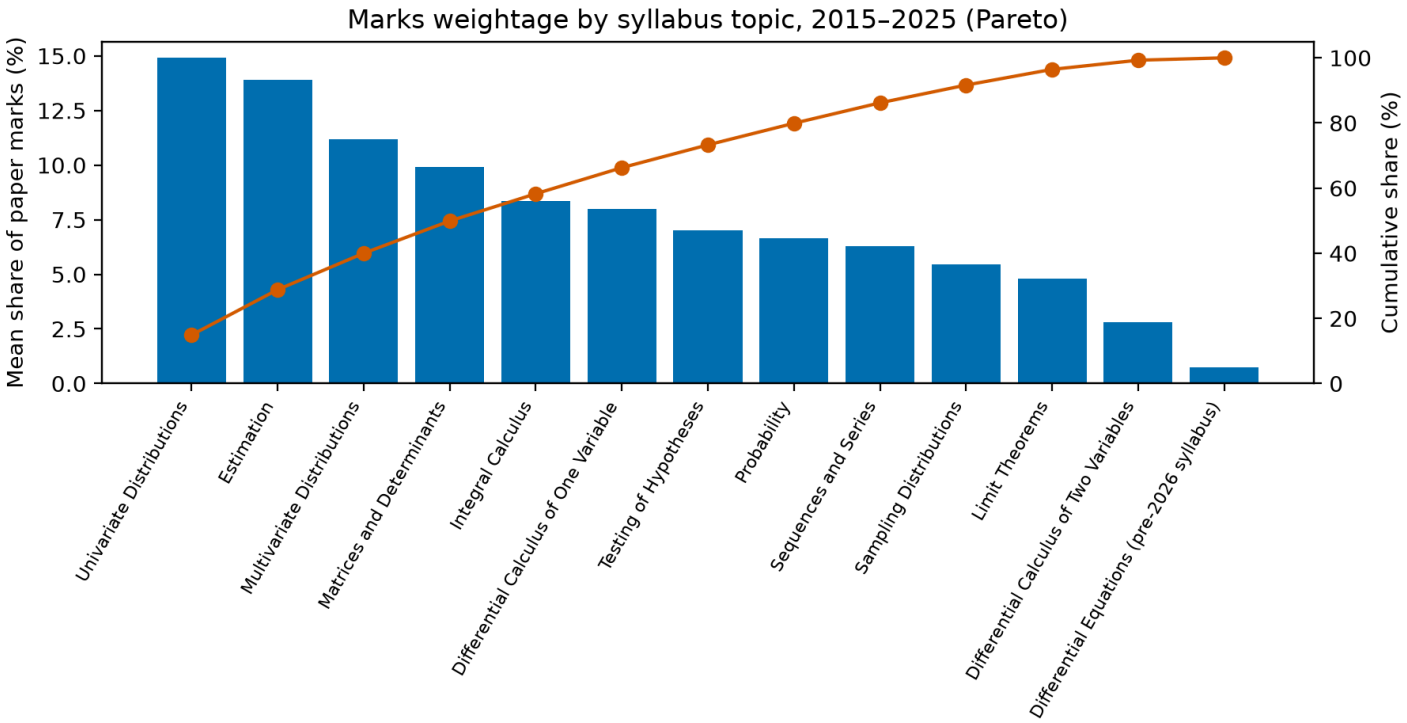


Source: JAM MS papers 2012–2025; top-level questions

3. Topic frequency and weightage

Marks share is the average over papers of (topic marks ÷ 100), so every paper counts equally. Frequency, weightage and consistency are separate columns.

Subject	Topic	Questions	Papers (of 11)	Mean marks share	Questions 2023–2025	Problem structures
Statistics	Univariate Distributions	103	11	14.9%	27	55
Statistics	Estimation	87	11	13.9%	26	40
Statistics	Multivariate Distributions	70	11	11.2%	23	41
Mathematics	Matrices and Determinants	66	11	9.9%	16	41
Mathematics	Integral Calculus	56	11	8.4%	9	22
Mathematics	Differential Calculus of One Variable	52	11	8.0%	12	32
Statistics	Testing of Hypotheses	45	11	7.0%	14	16
Statistics	Probability	45	11	6.6%	10	25
Mathematics	Sequences and Series	45	11	6.3%	10	20
Statistics	Sampling Distributions	37	10	5.5%	15	19
Statistics	Limit Theorems	33	11	4.8%	12	18
Mathematics	Differential Calculus of Two Variables	17	10	2.8%	6	5
Mathematics	Differential Equations (pre-2026 syllabus)	4	2	0.7%	0	1

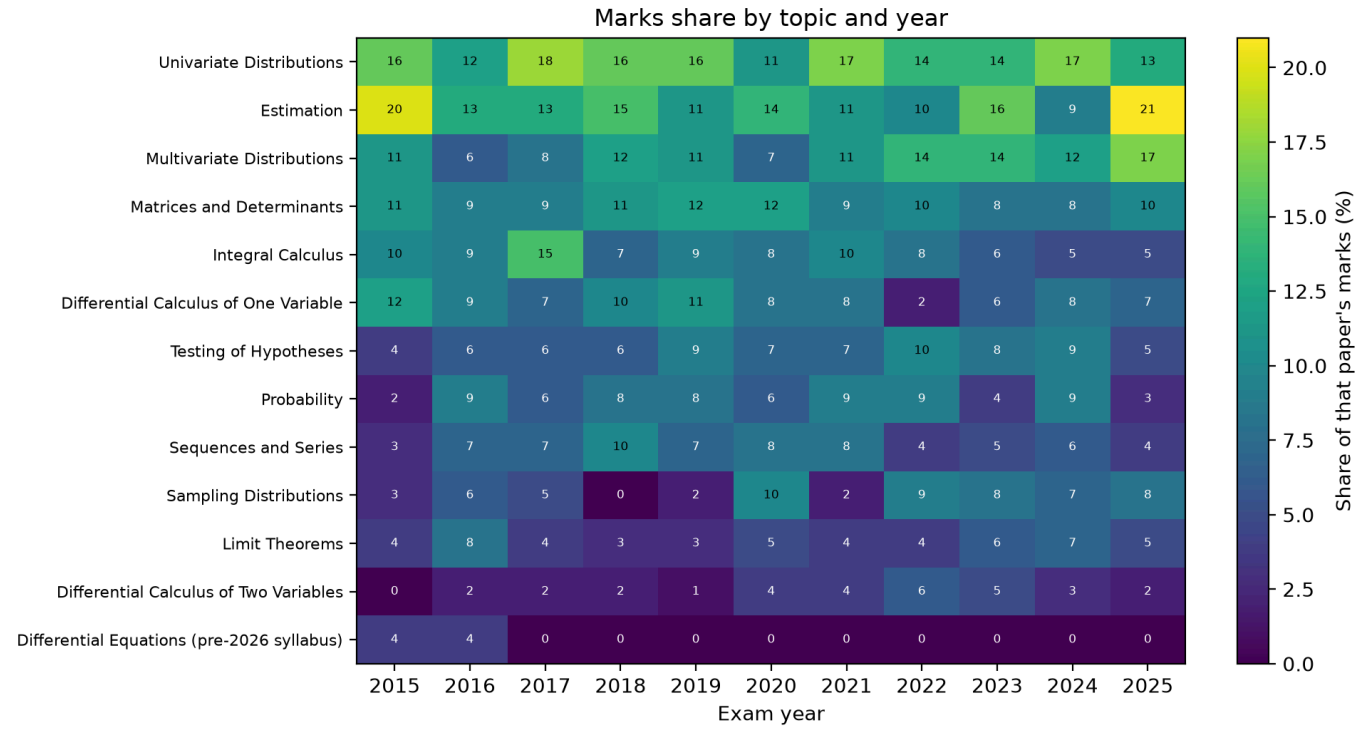


Source: JAM MS papers 2015–2025 (current format); n = 660 questions, 11 papers

4. Year-wise patterns

Average marks share, first four vs last three papers (descriptive only; with 11 papers no trend test is meaningful):

- Integral Calculus: 10.2% → 5.3%
- Differential Calculus of One Variable: 9.5% → 7.0%
- Differential Equations (pre-2026 syllabus): 2.0% → 0.0%
- Differential Calculus of Two Variables: 1.5% → 3.3%
- Sampling Distributions: 3.5% → 7.7%
- Multivariate Distributions: 9.2% → 14.3%



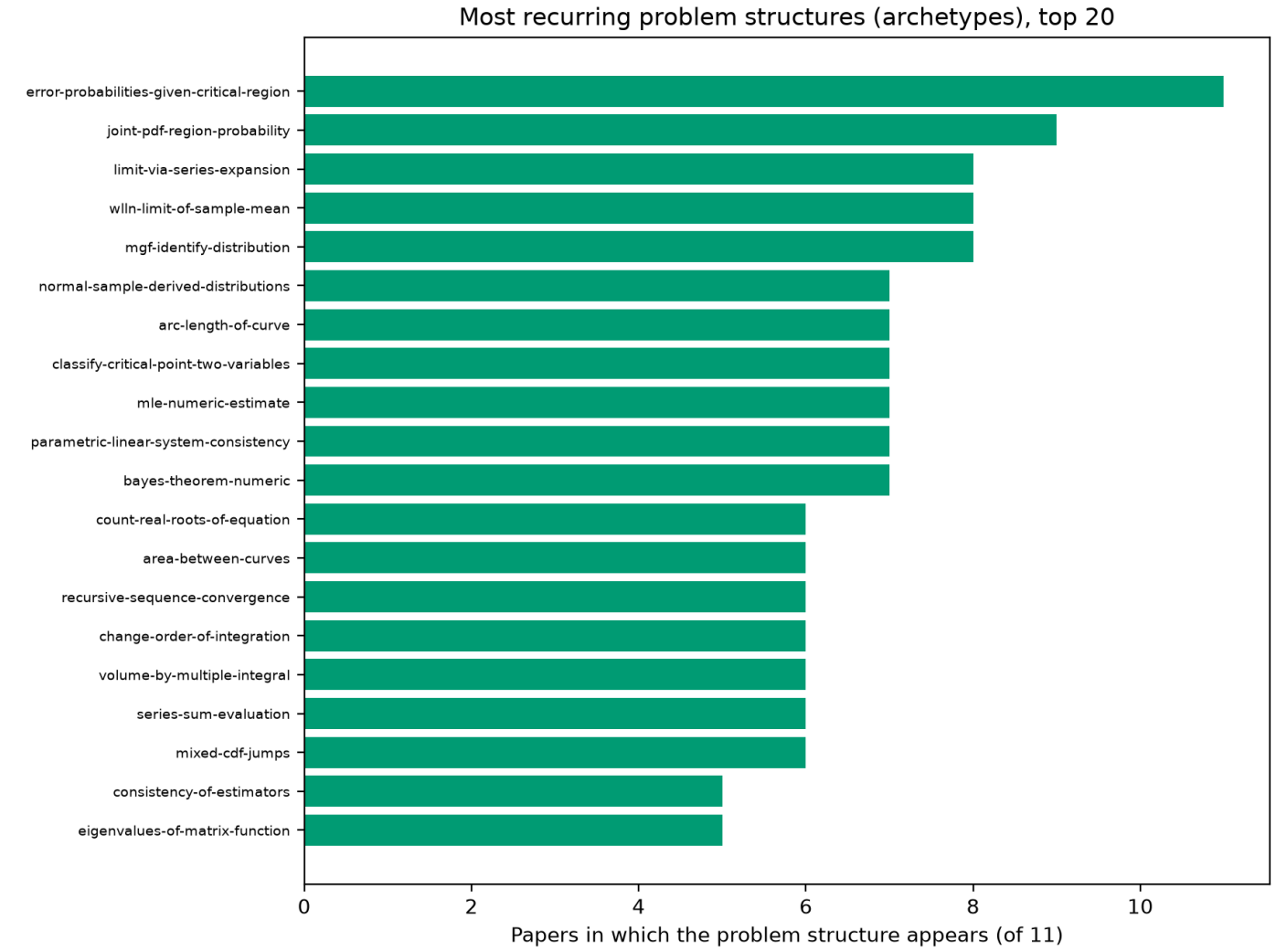
Source: JAM MS papers 2015–2025 (current format); n = 660 questions, 11 papers

5. Recurring problem structures

An archetype is a problem structure (e.g. "find error probabilities for a given critical region"). The most recurrent, by number of papers they appear in:

Topic	Archetype	Papers	Questions	Examples
Testing of Hypotheses	error-probabilities-given-critical-region	11	15	2015-MS-Q18, 2015-MS-Q3, 2016-MS-Q20, 2017-MS-Q21
Multivariate Distributions	joint-pdf-region-probability	9	10	2015-MS-Q12, 2016-MS-Q43, 2017-MS-Q15, 2019-MS-Q25
Differential Calculus of One Variable	limit-via-series-expansion	8	9	2016-MS-Q33, 2017-MS-Q27, 2019-MS-Q51, 2020-MS-Q13
Limit Theorems	wlln-limit-of-sample-mean	8	8	2016-MS-Q9, 2017-MS-Q17, 2018-MS-Q49, 2019-MS-Q8
Univariate Distributions	mgf-identify-distribution	8	8	2016-MS-Q5, 2017-MS-Q6, 2018-MS-Q20, 2020-MS-Q55
Sampling Distributions	normal-sample-derived-distributions	7	12	2016-MS-Q18, 2016-MS-Q41, 2017-MS-Q16, 2017-MS-Q46
Integral Calculus	arc-length-of-curve	7	8	2015-MS-Q57, 2016-MS-Q50, 2017-MS-Q30, 2017-MS-Q53
Differential Calculus of Two Variables	classify-critical-point-two-variables	7	7	2016-MS-Q27, 2017-MS-Q25, 2019-MS-Q2, 2020-MS-Q32
Estimation	mle-numeric-estimate	7	7	2017-MS-Q18, 2018-MS-Q8, 2019-MS-Q27, 2020-MS-Q49

Topic	Archetype	Papers	Questions	Examples
Matrices and Determinants	parametric-linear-system-consistency	7	7	2015-MS-Q58, 2016-MS-Q31, 2020-MS-Q4, 2021-MS-Q28



Source: JAM MS papers 2015–2025 (current format); n = 660 questions, 11 papers

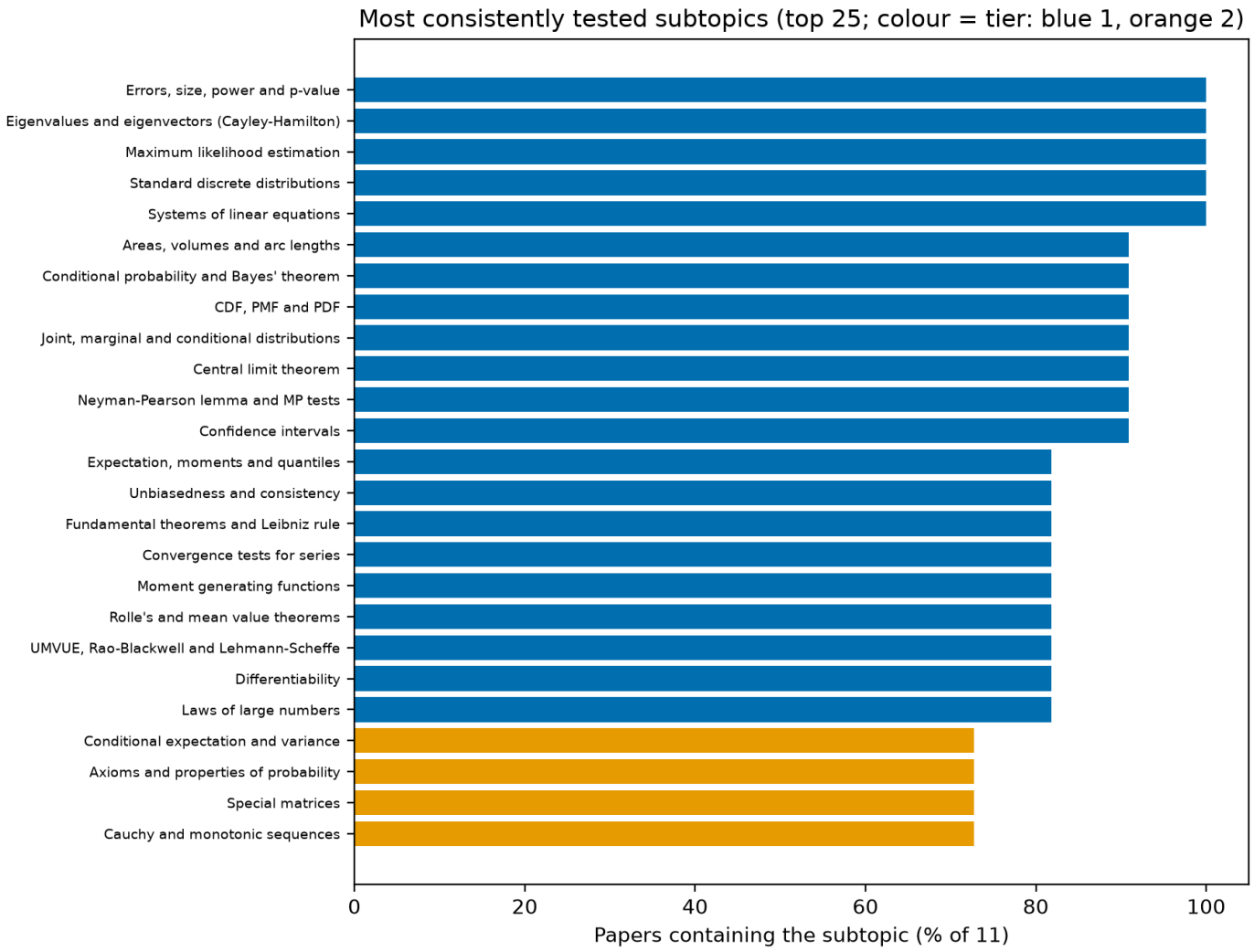
6. Priority tiers (subtopic level)

Rule, applied mechanically: **Tier 1** if the subtopic appears in ≥ 80% of papers or averages ≥ 2% of marks; **Tier 3** if it appears in < 40% of papers and averages < 1%; else Tier 2. Thresholds sit near the subtopic quartiles; they are a judgement, disclosed here. Counts: Tier 1 = 22, Tier 2 = 32, Tier 3 = 13, untested = 11. Tier 3 is "lower historical priority", never "skip".

Tier 1 subtopics

Topic	Subtopic	Questions	Papers (of 11)	Mean marks share
Integral Calculus	Areas, volumes and arc lengths	24	10	3.5%
Univariate Distributions	Expectation, moments and quantiles	23	9	3.4%

Topic	Subtopic	Questions	Papers (of 11)	Mean marks share
Testing of Hypotheses	Errors, size, power and p-value	22	11	3.4%
Matrices and Determinants	Eigenvalues and eigenvectors (Cayley-Hamilton)	22	11	3.3%
Univariate Distributions	CDF, PMF and PDF	20	10	3.1%
Probability	Conditional probability and Bayes' theorem	21	10	3.0%
Multivariate Distributions	Joint, marginal and conditional distributions	19	10	3.0%
Estimation	Maximum likelihood estimation	19	11	2.9%
Univariate Distributions	Standard discrete distributions	18	11	2.7%
Limit Theorems	Central limit theorem	17	10	2.6%
Estimation	Unbiasedness and consistency	17	9	2.6%
Integral Calculus	Fundamental theorems and Leibniz rule	16	9	2.4%
Sequences and Series	Convergence tests for series	14	9	2.4%
Matrices and Determinants	Systems of linear equations	14	11	2.4%
Multivariate Distributions	Covariance, correlation and joint moments	14	7	2.2%
Testing of Hypotheses	Neyman-Pearson lemma and MP tests	14	10	2.2%
Differential Calculus of One Variable	Rolle's and mean value theorems	12	9	2.0%
Estimation	UMVUE, Rao-Blackwell and Lehmann-Scheffe	12	9	1.9%
Estimation	Confidence intervals	11	10	1.8%
Univariate Distributions	Moment generating functions	13	9	1.7%
Differential Calculus of One Variable	Differentiability	12	9	1.6%
Limit Theorems	Laws of large numbers	11	9	1.5%



Source: JAM MS papers 2015–2025 (current format); n = 660 questions, 11 papers

Full tier list for all 78 subtopics: `outputs/tables/subtopics_2015_2025.csv`.

7. Potential future areas (hypotheses, not predictions)

- **Hypothesis** strong : Univariate Distributions and Estimation will stay the heaviest topics; they were heavy in every paper.
- **Hypothesis** moderate : Multivariate Distributions and Sampling Distributions keep the larger share they have had recently, and Integral Calculus stays lighter than in 2015–2018.
- **Hypothesis** weak : topics new to the 2026 syllabus (Descriptive Statistics, Nonparametric Methods, Stochastic Processes) may appear; there is no past data, so their likely weight is unknown.

Historical frequency is not a probability of appearance. No probabilities are given because 11 papers cannot support a defensible model.

8. Preparation recommendations

Suggested allocation (interpretation): 90% of study time in proportion to historical marks share, and a reserved 10% for the 11 untested 2026-syllabus subtopics. Don't drop any syllabus topic for low frequency.

Topic	Suggested study time
Univariate Distributions	13.5%
Estimation	12.6%
Multivariate Distributions	10.1%
Matrices and Determinants	9.0%
Integral Calculus	7.6%
Differential Calculus of One Variable	7.3%
Testing of Hypotheses	6.3%
Probability	6.0%
Sequences and Series	5.7%
Sampling Distributions	4.9%
Limit Theorems	4.4%
Differential Calculus of Two Variables	2.6%

- Work through past papers **by problem structure** (Section 5): 43.6% of questions reuse one.
- The paper is mostly numerical/application work (71.1%): practise timed computation, not proofs (proof-type questions: 1).
- Skip ODE-only preparation: not tested since 2016 and not in the 2026 syllabus.

9. Old-format papers (2012–2014)

2012 (25 questions, 300 marks, subjective parts), 2013 (30 questions: objective, fill-in-the-blank, descriptive) and 2014 (43 questions incl. 8 descriptive) differ in structure, so they are kept out of all shares above. Heaviest topics there:

Topic	Questions	Mean marks share
Estimation	13	16.0%
Univariate Distributions	15	14.7%
Multivariate Distributions	9	10.7%
Matrices and Determinants	10	8.7%
Probability	8	7.3%
Testing of Hypotheses	7	7.3%

10. Methodology and limitations

- Source: official papers from jam2026.iitb.ac.in (SHA-256 manifest). 4 papers are image scans, read with local OCR (PaddleOCR-VL); every page was audited against its image and about 70 OCR errors were fixed. Spot checks still found about 1 meaning-changing OCR slip per 4–5 unchecked pages.
- Each question is recorded as number, format, marks and pages, plus a one-line summary (no verbatim text). Counts and marks match each paper's printed instructions, section headers or printed answer key (2016, 2017 and 2022 totals are sums of printed section marks). The validator reports 0 errors on all 778 rows.
- Image verification: 2023 and 2024 in full; scans 2014, 2017 and 2025 PASS, but some of their pages were compared against the OCR text rather than the image. Text-based papers 2015–2022 were not sample-verified against images.
- Topic, archetype, cognitive type and repeat labels are model judgements (subjective). An independent reviewer agreed with 96% of uncertain topic calls (79/82). Archetypes and repeats are **moderate evidence at best**; cognitive type was not reviewed.
- The taxonomy is the 2026 syllabus, while older papers followed earlier syllabi. "Never tested" can mean "newly added".
- n = 11 papers: everything is descriptive, with no significance tests and no difficulty ratings (no score data).
- Reproducible: `python -m pyq.master && python -m pyq.report` regenerates every table, figure and this report from the frozen data. Code and data: github.com/aaditya19saini/OCR-QP-Tool.